

LOMASH RELIA

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EDUCATION

ABV-Indian Institute of Information Technology and Management

M.Tech. in Autonomous Systems and Machine Intelligence (CGPA Sem 1: 9.71/10)

Charotar University of Science and Technology

B.Tech. in Computer Engineering (CGPA: 9.41/10)

Gwalior, India

Aug. 2025 – Present

Anand, India

Sep. 2021 – Jun. 2025

TECHNICAL SKILLS

Languages: Python, C++, SQL, Dart, Verilog

AI & Vision: LLMs, VLMs, VLA, Computer Vision, GANs, Model Quantization, NLP

Robotics: MuJoCo, ROS2, Control Systems, Inverse Kinematics, Visual Servoing (IBVS), LeRobot, Edge AI

Frameworks: PyTorch, TensorFlow, FastAPI, LangChain, LlamaIndex, OpenCV, Flutter, Selenium

Tools & Hardware: Linux, Git, Raspberry Pi, FPGA (Basys3/Artix-7), PostgreSQL, Firebase, VNC

EXPERIENCE

Graduate Researcher (MTP Student)

ABV-IIITM Gwalior (Supervisor: Dr. Somesh Kumar)

- Developing mathematical control systems for the SO-101 robotic arm using MuJoCo simulations.
- Researching Vision-Language-Action (VLA) models for autonomous planetary robotic manipulation.

Aug. 2025 – Present

Gwalior, India

Research Intern

Space Applications Center, ISRO

- Architected a planetary rover prototype with Raspberry Pi featuring monocular depth estimation, stereo photogrammetry lunar simulation, and modular Bluetooth control architecture.
- Quantized and optimized deep learning models for edge deployment, achieving near real-time inference.

Jan. 2025 – Jun. 2025

Ahmedabad, India

Research Collaborator

Indian Institute of Tropical Meteorology

- Engineered a deep learning pipeline for 4-channel Tropical Cyclone Image-to-Intensity Regression.
- Analyzed neural network architectures to improve intensity estimation accuracy from satellite imagery.

Aug. 2024 – Nov. 2024

Pune, India

Full Stack Engineer

Capitalize

- Developed Flutter applications integrated with FastAPI backends and RESTful API structures.
- Automated data collection pipelines using Selenium and rotating proxies, ensuring a 99% success rate.

May 2024 – Jul. 2024

Remote

PROJECTS

Eye-in-Hand IBVS for SO-101 Arm | MuJoCo, OpenCV, Python

- Implemented Image-Based Visual Servoing (IBVS) using ArUco detection and Chaumette's pixel-based control, achieving reliable convergence to 30–40 px feature error.
- Integrated joint/velocity limits, automatic divergence recovery, and a JSON-based teaching tool.

Dec. 2025 – Jan. 2026

SIFT Feature Detector on FPGA | Verilog, Vivado, Basys3 (Artix-7)

- Designed a hardware Difference of Gaussian (DoG) pipeline in Verilog on the Artix-7 FPGA, using separable 2D convolution with 5-tap shift registers and Block RAM frame buffers to process 128×128 grayscale images.
- Optimized fixed-point arithmetic via bit-shifting to minimize DSP slice usage; ran parallel dual- σ Gaussian chains with edge mirroring, verified via Python hex tooling and Vivado testbenches.

Dec. 2025 – Present

AI-Powered Planetary Rover | PyTorch, OpenCV, Raspberry Pi

- Deployed object detection and monocular depth estimation on Bookworm OS with a 65°C thermal cap.
- Designed dual navigation modes supporting autonomous movement and remote manual control via VNC.

Jan. 2025 – Jul. 2025

LLM-Based Knowledge Extraction System | FastAPI, LangChain, RAG

- Built a RAG pipeline for text-to-knowledge transformation using LLMs and LangChain.

Apr. 2024

ACHIEVEMENTS & LEADERSHIP

- **IEEE Leadership:** Student Branch Secretary, IEEE MTTs; selected as an **IEEE Space 2026 Ambassador** to represent IEEE space initiatives and promote advancements in space technology.
- **Teaching Experience:** Graduate Teaching Assistant, ABV-IIITM Gwalior.
- **Academic Rankings:** KEA PG CET 2025 - Rank 52; GATE 2025 (Data Science & AI) - AIR 4687.
- **Publications:** Research contributions published in the *WGNE Blue Book* (2024) and a *Springer* research collection (2024).
- **Conference Presentations:** Presented and published work at **IEEE InGARSS 2025**, **PRL Venus 2024**, and **ICHCS 2024**.